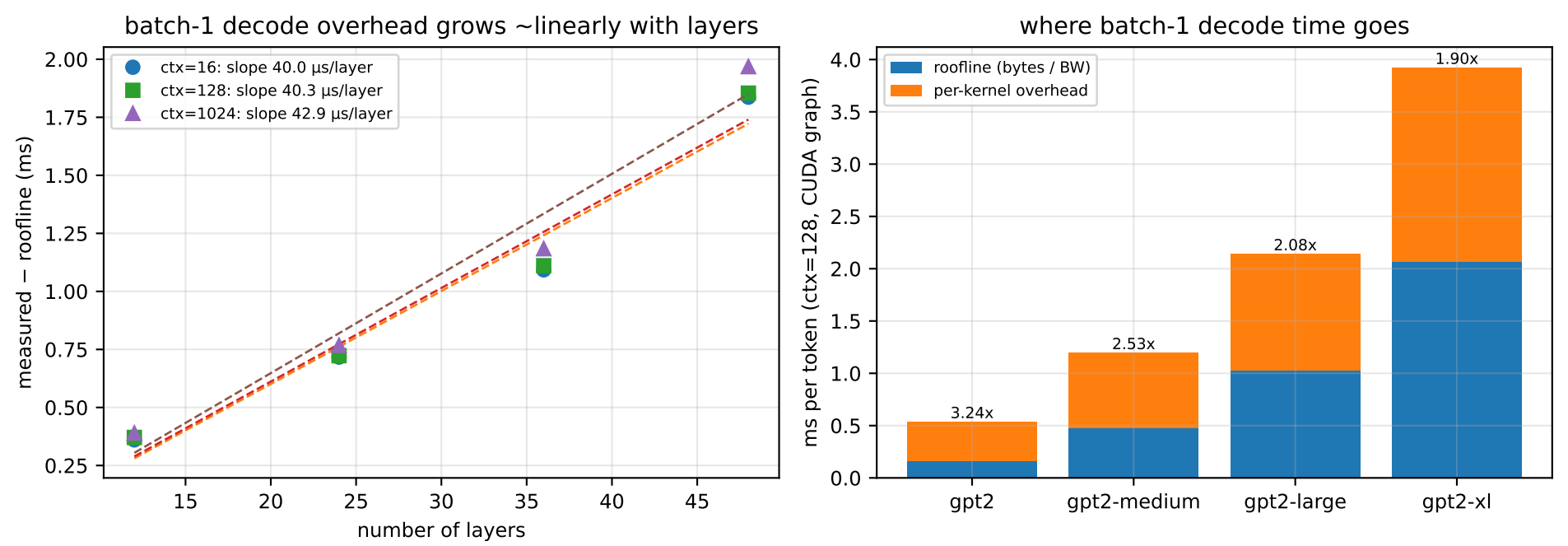
batch-1 decode overhead grows ~linearly with layers
ctx=16: slope 40.0 µs/layer
ctx=128: slope 40.3 µs/layer
ctx=1024: slope 42.9 µs/layer
measured − roofline (ms)
number of layers
where batch-1 decode time goes
roofline (bytes / BW)
per-kernel overhead
ms per token (ctx=128, CUDA graph)
3.24x
2.53x
2.08x
1.90x
gpt2
gpt2-medium
gpt2-large
gpt2-xl